Shinu Rathod

Data Scientist · AI Researcher · ML Engineer

AI Researcher at IIT Ropar | Cybersecurity AI Division

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TECHNICAL SKILLS

- **Programming**
  Python · SQL · Kotlin · R
- **ML / DL Frameworks**
  PyTorch · TensorFlow · Keras · Scikit-learn · Hugging Face · LoRA / QLoRA · ONNX
- **Generative AI**
  LLM Fine-Tuning · Prompt Eng. · RAG · LangChain · LlamaIndex · Pinecone · ChromaDB
- **Agentic AI** (Upskilling)
  AI Agents · ReAct · Tool Use · Multi-Agent
- **Deployment & MLOps**
  Flask · FastAPI · Docker · Streamlit · Android/ONNX
- **Visualization**
  Power BI · Matplotlib · Seaborn

AI TOOLS

Claude · ChatGPT · Gemini · Perplexity · Claude Code

EDUCATION

Data Science Trainee
360DigiTMG, Bengaluru
Feb 2023 – Feb 2024

Professional Training Program
Jitendra Mishra Academy (JMA)
2022 – 2023

Bachelor of Computer Applications
Karnataka College, Bidar
2018 – 2021
CGPA: 7.64

CERTIFICATIONS

Data Science & Analytics Internship
Zidio Development · Dec 2024
bit.ly/zidio-cert

Autism Detection Internship
AiSPRY · May 2025
bit.ly/aispry-cert

PROFESSIONAL SUMMARY

AI Researcher at **IIT Ropar** and Data Scientist with proven expertise in building production-grade ML systems. Currently leading research on **on-device phishing URL detection** using Transformer fine-tuning (MiniLM + LoRA) trained on **35.3M URLs**, achieving **98.91% AUC-ROC** and deployed as an Android app via ONNX INT8 quantization. Skilled in Generative AI — RAG pipelines, LLM fine-tuning, and multi-agent orchestration. Actively upskilling in Agentic AI frameworks. Seeking to leverage deep AI expertise as an **AI/ML Engineer or Data Scientist** at a product-driven organization.

WORK EXPERIENCE

AI Researcher — Cybersecurity AI Division
2025 – Present

IIT Ropar RESEARCH

- Fine-tuned **MiniLM-L12 Transformer + LoRA** on **35.3M URLs** for phishing detection; achieved **AUC-ROC 98.91%**, FNR ≤10%, FPR 2.47% on 4.4M-URL test set.
- Designed 6-stage canonical URL preprocessing pipeline handling Unicode homoglyphs, IDNA 2008, recursive percent-decoding (10-pass), and blob masking across **40M+ URLs** using **12-core multiprocessing**.
- Deployed production model as **PhishGuard Android App** — ONNX INT8 quantization reducing model to **32.6 MB** with **<100ms on-device inference**, zero internet required.
- Validated cross-platform parity via two-stage golden vector test (bit-exact token IDs + probability ε=0.001).

PyTorch · Transformers · LoRA · ONNX Runtime · Kotlin · Android · Jetpack Compose · Focal Loss

Data Scientist Intern
Apr 2025 – May 2025

AISPRY, Hyderabad, Telangana

- Built AI-powered **Autism Spectrum Detection system** using TensorFlow Lite + MediaPipe pose estimation; achieved **91.8% accuracy** on clinical video/webcam input.
- Integrated NLP-based multimodal analysis with Gemini-powered chatbot; deployed via React + Flask + Docker, reducing preprocessing time by **15%**.

TensorFlow Lite · MediaPipe · React · Flask · Docker · Gemini API

Data Science & Analytics Intern
Aug 2024 – Nov 2024

Zidio Development, Bengaluru, Karnataka

- Engineered AI-driven solutions using Python and TensorFlow, improving system efficiency by **15%**.
- Built and deployed scalable ML models; created Power BI dashboards supporting strategic decision-making.
- Automated workflows using Python + SQL, enhancing operational efficiency by **25%**.

Python · TensorFlow · Power BI · SQL

Data Science Trainee
Feb 2023 – Feb 2024

360DigiTMG, Hyderabad, Telangana

- Built predictive maintenance model with Scikit-learn, reducing machine downtime by **20%**.
- Developed behavior detection system for AI Tutor platform, improving user engagement by **10%**.

Scikit-learn · Python · SQL · Power BI

KEY PROJECTS

PhishGuard — On-Device Phishing URL Detection (IIT Ropar)

github.com/SHINU4RATHOD/PhishDetect

★ AUC-ROC 98.91% · 35.3M URLs · 32.6 MB Android app · <100ms inference

- Transformer (MiniLM + LoRA) + Focal Loss trained with FP16 AMP on 26.5M URLs; INT8 ONNX quantized for mobile.

PyTorch | LoRA | ONNX | Kotlin | Android

Medical Chatbot + Multi-PDF RAG System

GitHub ↗

★ 30% faster data retrieval · LangChain + Pinecone + ChromaDB

- RAG pipeline with vector database for medical Q&A with source attribution; MultiPDF reader with Google Gemini.

LangChain | LLMs | Pinecone | ChromaDB | RAG

AI-Powered Code Analyzer

GitHub ↗

★ 25% improvement in code review efficiency

- LangChain + ChromaDB semantic code search; intelligent insights and recommendations via LLM reasoning.

LangChain | ChromaDB | LLMs | Python